SS-C-181e

SEPTEMBER 23, 1960

SUPERSEDING
Int. Fed. Spec. SS–C–00181d (COM–NBS)
November 18, 1959 and
Fed. Spec. SS–C–181c
April 28, 1954

FEDERAL SPECIFICATION

CEMENT, MASONRY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types.—This specification covers two types of masonry cement, as follows:

Type I.—This type of masonry cement is intended for use in masonry construction above grade where not exposed to frost action.

Type II.—This type of masonry cement is intended for general use where mortars for masonry are required.

Type II masonry cement shall be furnished unless otherwise specified.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

~~UU–P–236~~ NNN-P-141 73 — Paper, Filter, Analytical.

UU–S–48 — Sacks, Shipping, Paper.

Federal Standards:

Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 158 — Cement, Hydraulic; Sampling, Inspection and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–129 Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The manufacturer is given a wide range in selection of materials and processes of manufacture in order that cement of the prescribed quality may be produced.

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3.2 Water repellency. — The masonry cement when required to be water-repellent shall comply with the requirements of the one-hour absorption test (see 4.4).

3.3 Nonstaining. — The masonry cement when required to be nonstaining to limestone shall comply with the requirements for soluble "alkali" (see 4.5 and 6.3).

3.4 Detail requirements.—Masonry cement of each of the two types shall conform to the respective requirements prescribed in Table I, when tested in accordance with the methods outlined in Section 4.

3.5 The inspector or agency submitting the sample shall report the net weight in pounds per bag of masonry cement packaged by the manufacturer, as part of the information furnished with the samples.

3.6 Cause for rejection.—The cement shall be rejected if it fails to meet any of the requirements of this specification. Cement remaining in storage prior to shipment for a period longer than 6 months after test may be retested, and shall be rejected if it fails to meet any of the requirements of this specification.

TABLE I. — *Detail requirements when tested as shown in 4.1 to 4.5, inclusive*

Test	Type I	Type II
Fineness: Residue No. 325 sieve, maximum percent	15	15
Time of setting, Gillmore method:		
Initial set, hours, minimum	2	2
Final set, hours, maximum	24	24
Soundness, autoclave expansion: Maximum expansion, percent	1.0	1.0
Comprehensive strength[1] (average of 3 cubes:		
7 day, minimum, p.s.i.	250	500
28 day, minimum, p.s.i.	500	900
Air-entrainment: Minimum percent by volume	12	12
Flow after suction: Minimum, percent	70	70
Absorption[2] (Average increase in weight of cubes in 1 hour), grams, maximum	5.0	5.0
Staining[3] Soluble "Alkali", calculated as Na_2O maximum percent	0.03	0.03

[1] Strength at 28 days shall be greater than at 7 days.

[2] Applies only when a water-repellent cement is specifically requested by the purchaser in the invitation for bids (see 4.4 and 6.4).

[3] Applies only when a nonstaining cement is specifically requested by the purchaser in the invitation for bids (see 4.5 and 6.3).

4. METHODS OF SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The applicable methods outlined in Federal Test Method Standard No. 158, of the issue in effect on date of invitation for bids, form a part of this specification and shall be followed except where specific changes are prescribed in the following paragraphs. The methods listed below refer to the methods found in Federal Test Method Standard No. 158.

4.2.1 *Fineness.* — The residue on the No. 325 (44 micron) sieve shall be determined as directed in method 2111.

4.2.2 *Preparation of cement pastes.* — Cement paste may be mixed by the procedure in method 2001, paragraph 3.10.7 (Machine Mixing) or by the following procedure: Five

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hundred grams of the cement shall be placed in the form of a crater on a plane nonabsorbing plate. Part of the quantity of mixing water required for normal consistency (par. 4.2.3) shall be poured into the crater, the cement mixed into the water wtih the finger tips until all of the water is taken up by the cement, the remainder of the water added, and the paste then mixed with the fingers until all the water is absorbed. The mixing operation shall then be completed by continuous vigorous squeezing and kneading the paste with the hands for 3 minutes. During the operation of mixing, the hands shall be protected by rubber gloves.

4.2.3 *Normal consistency and time of setting.* — Normal consistency shall be determined by the Vicat apparatus in accordance with par. 3.2 method 2201, and the time of setting shall be determined in accordance with par. 3 of method 2231, except that the mixing of the paste shall be performed in accordance with par. 4.2.2 of this specification.

4.2.4 *Soundness.* — Soundness shall be determined in accordance with method 2211, except that the paste shall be mixed according to par. 4.2.2 of this specification and that the specimens shall be removed from the molds at 48 hours ± ½ hour for measurement and testing.

4.2.5 *Specific gravity.* — The specific gravity of the masonry cement shall be determined in accordance with method 2601, using kerosene as the liquid. The specific gravity so determined is used in the calculation of the air content of the mortars, method 2401.

4.2.6 *Mortars.*

4.2.6.1 *Proportioning and consistency.* — Mortars for compressive strength, flow after suction, absorption, and air-entrainment (Note) shall be proportioned to contain the weight of cement, in grams, equal to six times the printed bag weight in pounds, and 1440 grams of sand. The sand shall consist of 720 grams of graded Ottawa sand and 720 grams of 20-30 Ottawa sand. The quantity of water, measured in milliliters, shall be that required to produce a flow of 110 ± 5 as determined with 25 drops of the flow table.

Note.—The specified mortar proportions are approximately the 1:3 nominal proportions, by volume, commonly specified for construction, as indicated in the following calculation. If a bag of masonry cement contained 1 cu. ft. of cement and 1 cu. ft. of loose damp sand contained 80 pounds of dry sand:

$$C = \frac{1440}{240} \times \text{weight per bag (lbs.)} = 6 \times \text{weight per bag (lbs.)}$$

where C is the number of grams of cement to be used in the mortar with 1440 grams of sand, and 240 is the weight in pounds of dry sand in 3 cu. ft. of loose damp sand.

4.2.6.2 *Mixing and determination of flow.* — Mortars shall be mixed using the apparatus and procedure given in method 2001, par. 3.10. The flow shall be determined in accordance with method 2301, par. 4.3.

4.2.6.3 *Air-entrainment.* — If the mortar has the correct flow, another portion of the mortar shall be used for determination of the entrained air. The air content shall be determined in accordance with pars. 3.3 and 4.1 of method 2401. The air content shall be calculated using the general formula for cements other than portland cement.

4.2.6.4 *Compressive strength.* — After determination of the flow and the entrained air, the mortar shall be returned to the bowl and remixed 15 seconds at the medium speed. Cubes for compressive strength shall then be molded in accordance with par. 5.1 of method 2301.

4.2.6.5 *Storage.* — All test specimens, immediately after molding, shall be stored in the moist closet for 48 hours. The cubes shall then be removed from the molds and placed in the moist closet for 5 days in such a manner as to allow free circulation around 5 faces of the specimens. At the age of 7 days the cubes for 28 day tests shall be placed in water in the storage tank. The remainder of the testing procedure shall conform to the appropriate paragraphs of method 2301.

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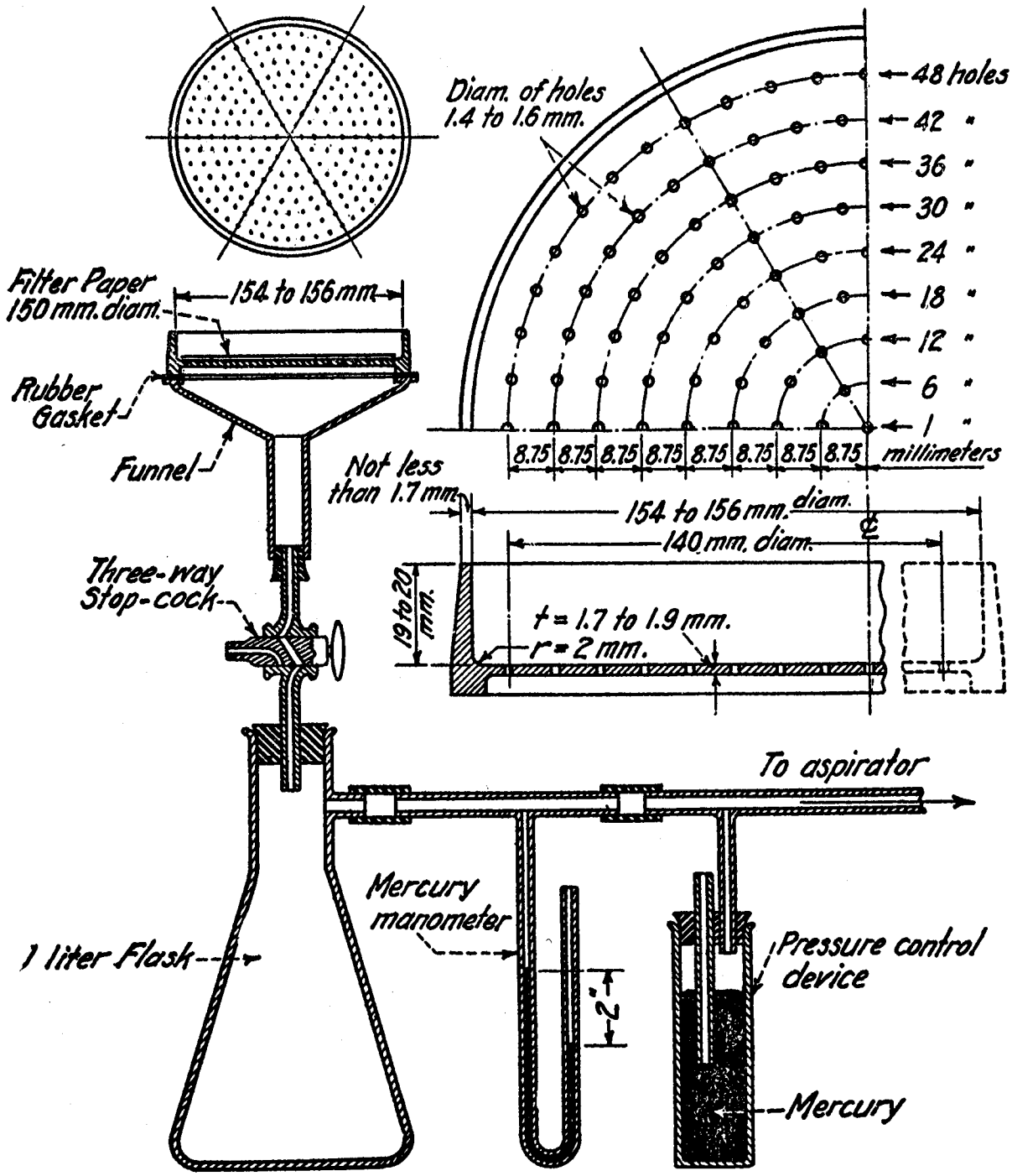


Figure 1.

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4.3 Flow after suction.

4.3.1 *Apparatus.*—For the flow after suction test an apparatus essentially the same as that shown in figure 1 shall be used. This apparatus consists of a water aspirator controlled by a mercury column relief and connected by way of a three-way stopcock to a funnel upon which rests a perforated dish. The perforated dish shall be made of metal not attacked by masonry mortar. The metal in the base of the dish shall have a thickness of 1.7 to 1.9 mm. and shall conform with the outline shown in figure 1. The bore of the stopcock shall have a 4-mm. diameter, and the connecting glass tubing shall have a minimum inside diameter of 4 mm. A mercury manometer, connected as shown in figure 1, indicates the vacuum. A synthetic rubber gasket shall be permanently sealed to the top of the funnel and shall be lightly coated with petrolatum or light cup grease during a test to insure a seal between the funnel and dish. Care should be taken to insure that none of the holes in the perforated dish are clogged from the grease used on the rubber gasket. Hardened filter paper equivalent to type I, class 4, Federal Specification UU–P–236 shall be used. It shall be of such diameter that it will lie flat and completely cover the bottom of the dish. Some other source of vacuum may be used, provided it will furnish a uniform pressure difference equal to two inches of mercury.

4.3.2 *Procedure.* — Adjust the mercury relief column so as to maintain a vacuum of 2 in. as measured on the manometer. Seat the perforated dish on the greased gasket of the funnel. Place a wetted filter paper in the bottom of the dish. Turn the stopcock to apply the vacuum to the funnel and check the apparatus for leaks and to determine that the required suction is obtained. Then turn the stopcock to shut off the vacuum from the funnel. Prepare the test mortar in accordance with the procedure described in 4.2. Immediately after making the flow test return the mortar on the flow table to the mixing bowl and remix the entire batch for 15 seconds at the medium speed. Immediately after remixing the mortar, fill the perforated dish with the mortar to slightly above the rim. Tamp the mortar 15 times with the tamper. Ten of the tamping strokes shall be applied at approximately uniform spacing adjacent to the rim of the dish and with the long axis of the tamping face held at right angles to the radius of the dish. The remaining 5 tamping strokes shall be applied at random points distributed over the central area of the dish. The tamping pressure shall be just sufficient to insure filling of the dish. On completion of the tamping, the top of the mortar should extend slightly above the rim of the dish. Smooth off the mortar by drawing the flat side of the straightedge (with the leading edge slightly raised) across the top of the dish. Then cut off the mortar to a plane surface flush with the rim of the dish by drawing the straightedge with a sawing motion across the top of the dish, in two cutting strokes, starting each cut from near the center of the dish. If the mortar is pulled away from the side of the dish during the cutting, gently press the mortar back into contact, using the tamper. Turn the stopcock to apply the vacuum to the funnel. The time elapsed between adding the cement to the water, and the time of applying the vacuum, shall not exceed 8 minutes. After suction for 60 seconds quickly turn the stopcock to expose the funnel to atmospheric pressure. Immediately slide the perforated dish off the tunnel, touch it momentarily on a damp cloth to remove droplets of water and set the dish on the table. Then using the rubber bowl-scraper, plow and mix the mortar in the dish for 15 seconds. Upon completion of the mixing, fill the flow-table mold with the mortar and redetermine the flow. The entire operation shall be carried out without interruption and as quickly as possible, and shall be completed within an elapsed time of 11 minutes after adding the cement to the water.

4.3.3 *Calculation.* — Calculate the flow-after-suction value for the mortar as follows:

$$\text{Flow-after-suction value} = \frac{A}{B} \times 100,$$

where

A = flow after suction, and

B = flow immediately after mixing

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4.4 Absorption test for water repellency.—Mortar for the cubes used in the water absorption test shall be proportioned and mixed as directed in 4.2. After mixing the mortar the cubes shall be tamped in the usual manner in cube molds that have not been greased or oiled. After 48 hours storage in the moist cabinet, they shall be removed from the molds and placed in the moist cabinet for 5 days in such a manner as to allow free circulation of air around at least 5 faces of the specimens. At the age of 7 days, they shall be placed in a drying oven maintained at a temperature of 105°–110° C. for 20 to 24 hours, then removed, placed in air at 23° ± 3° C. for 2 hours, weighed to the nearest 0.1 gram, and then placed in water at 23° ± 1.7° C. to a depth of ¼ inch with the top side as cast placed downward. At the end of 1 hour, the specimens shall be removed, drained for 5 minutes, wiped with a damp cloth, and again weighed to the nearest 0.1 gram. The average gain in weight of three specimens shall be reported as the 1-hour absorption (see 3.2).

4.5 Staining test.—The determination of soluble alkali shall be made as follows: To 150 grams of sample in a 400-ml. beaker, add 250 ml. distilled water. Stir thoroughly, let stand 30 minutes at room temperature, and stir again. Filter through a Buchner funnel, transfer the insoluble matter to the original beaker and rinse the funnel with 150 ml. distilled water into the beaker containing the insoluble matter. Stir thoroughly, let stand 30 minutes at room temperature, and stir again. Filter as above and again return the insoluble matter to the original beaker. Wash the funnel with 100 ml. distilled water into the beaker containing the insoluble matter. Stir thoroughly, let stand 30 minutes at room temperature, and stir again. Filter as above. Acidify the filtrate with 5 ml. of concentrated HCl. Stir until cloudiness disappears. Evaporate the solution to about 400 ml. Cool to room temperature and transfer quantitatively to a 500-ml. volumetric flask. Dilute to 500 ml. Transfer a 50-ml. aliquot of this solution to a 100-ml. volumetric flask. Refer to method 1401, determination of sodium and potassium oxides in portland cement by flame photometer. Pour 9.0 ml. (Note) of the stock $CaCl_2$ solution, (63,000 ppm CaO), into the 100-ml. flask, and dilute the solution to 100 ml. Determine the sodium and potassium-oxide content of this solution as directed in method 1401. Read the parts per million of each in the solution from the calibration curve.

Calculation:

$$\text{Percent soluble } Na_2O \text{ in cement} = \frac{\text{ppm } Na_2O \text{ in solution in 100-ml. flask}}{1500}$$

$$\text{Percent soluble } K_2O \text{ in cement} = \frac{\text{ppm } K_2O \text{ in solution in 100-ml. flask}}{1500}$$

Then percent $K_2O \times 0.658$ = percent Na_2O equivalent of K_2O, and percent Na_2O + (percent Na_2O equivalent of K_2O) = soluble alkali calculated as Na_2O.

If so much alkali is present that the value cannot be read on the standard curve, a more dilute solution may be prepared from another aliquot which will bring the readings for the flame photometric measurement within the range of the curve. Appropriate changes in the calculations must then be made.

Note.—If no calcium were dissolved from the cement, 10 ml. of the stock $CaCl_2$ solution would be required for each 100 ml. of solution tested. It has been found that about 800 ppm. of calcium as CaO are furnished by the cement under the conditions of this test.

4.6 Inspection of preparation for delivery.—The packing and marking of the cement shall be examined to determine compliance with the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Unless otherwise specified, the cement shall be delivered in bags with the words masonry cement (or mortar cement), the brand, and name of the manufacturer, and the net weight in pounds plainly marked

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thereon. All packages shall be in good condition at the time of inspection. Packages varying more than 5 percent from the marked weight may be rejected, and if the average weight of packages in any shipment as shown by weighing 50 packages taken at random is less than that marked, the entire lot represented shall be rejected.

5.2 **Packing.**—The packing of the cement shall be level A, B, or C as specified (see 6.2). Unless otherwise specified (see 6.2), the cement shall be packed in quantities of 70 pounds.

5.2.1 *Level A.* — Cement shall be packed in multiwall sacks conforming to Federal Specification UU–S–48, X series.

5.2.2 *Level B.* — Cement shall be packed in multiwall sacks conforming to Federal Specification UU–S–48, L/W series.

5.2.3 *Level C.* — Cement shall be packed in a manner which will assure arrival at destination in satisfactory condition and which will be acceptable to the carrier at the lowest transportation cost.

5.3 **Marking.**—In addition to any special marking required by the contract or order, the cement shipments shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.**—The cement covered by this specification is intended for use in masonry mortars.

6.2 **Ordering data.**—Unless type I is specifically ordered by the purchaser, type II masonry cement shall be furnished. The requirement for nonstaining cement and/or water repellent cement apply only when specified by the purchaser. Purchasers should exercise any desired options offered herein (see 3.2, 3.3, and 5.1).

6.3 **Nonstaining.**—The amount and nature of the staining material in limestones seems to vary with the stone. The alkali in any cement may, therefore, induce markedly different staining on different stone, even though the stone may have come apparently from the same source. The amount of water-soluble alkali permitted by the specification should not cause stain unless stone high in staining material has been used, or unless insufficient means have been used to prevent infiltration of water into the masonry.

6.4 Water-repellent cement meeting the requirements of the 1-hour absorption test is intended for special use to retard the passage of water through masonry mortar.

6.5 It is believed that this specification adequately describes the qualities necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.6 **Storage.**—The cement should be stored in a suitable weather-tight building which will protect it from dampness, and in such a manner as to permit safe and easy access for proper inspection and identification of each lot.

6.7 **Transportation description.** — Transportation description applicable to this item is:

Cement, Masonry

Carload minimum weight 50,000 pounds.

Motor volume minimum weight 50,000 pounds.

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MILITARY INTERESTS:
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